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| **Started** | Tuesday, 20 October 2026, 3:34 PM |
| **Completed** | Tuesday, 20 October 2026, 3:40 PM |
| **Duration** | 6 mins 15 secs |
| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By comparing each element sequentially
- b. By dividing the list length by two ✓
- c. By using a hash function
- d. By starting from the first element

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It does not work with negative numbers
- d. It can only be applied to large lists

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By using a hash function
- c. By dividing the list into halves
- d. By sorting the list first

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the left sublist ✓
c. In the entire list
d. In the right sublist

Question **5**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Sorting elements in a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Linear search ✓
c. Binary search
d. Hash search

Question **7**

Correct

Mark 1.00 out of 1.00

**_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.**

a. Complexity

b. Insertion

c. Selection

d. Bubble ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 89 and 94

c. 90 and 99 ✓

d. 89 and 99

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It compares elements sequentially

c. It can be applied only if the list is sorted ✓

d. It works on unsorted lists

Question **10**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search

b. Linear search

c. Binary search ✓

d. Hash search

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element

b. The middle element is chosen randomly

c. The search stops

d. The lower middle element is chosen as the middle element ✗

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **13**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Hash search
c. Linear search ✓
d. Binary search

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Must use a sorted array
c. There must be a mechanism to access middle element directly
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O🥱
c. O(log n)
d. O(1) ✓